

Scientific Calculator Project

Python • Tkinter • GUI Application

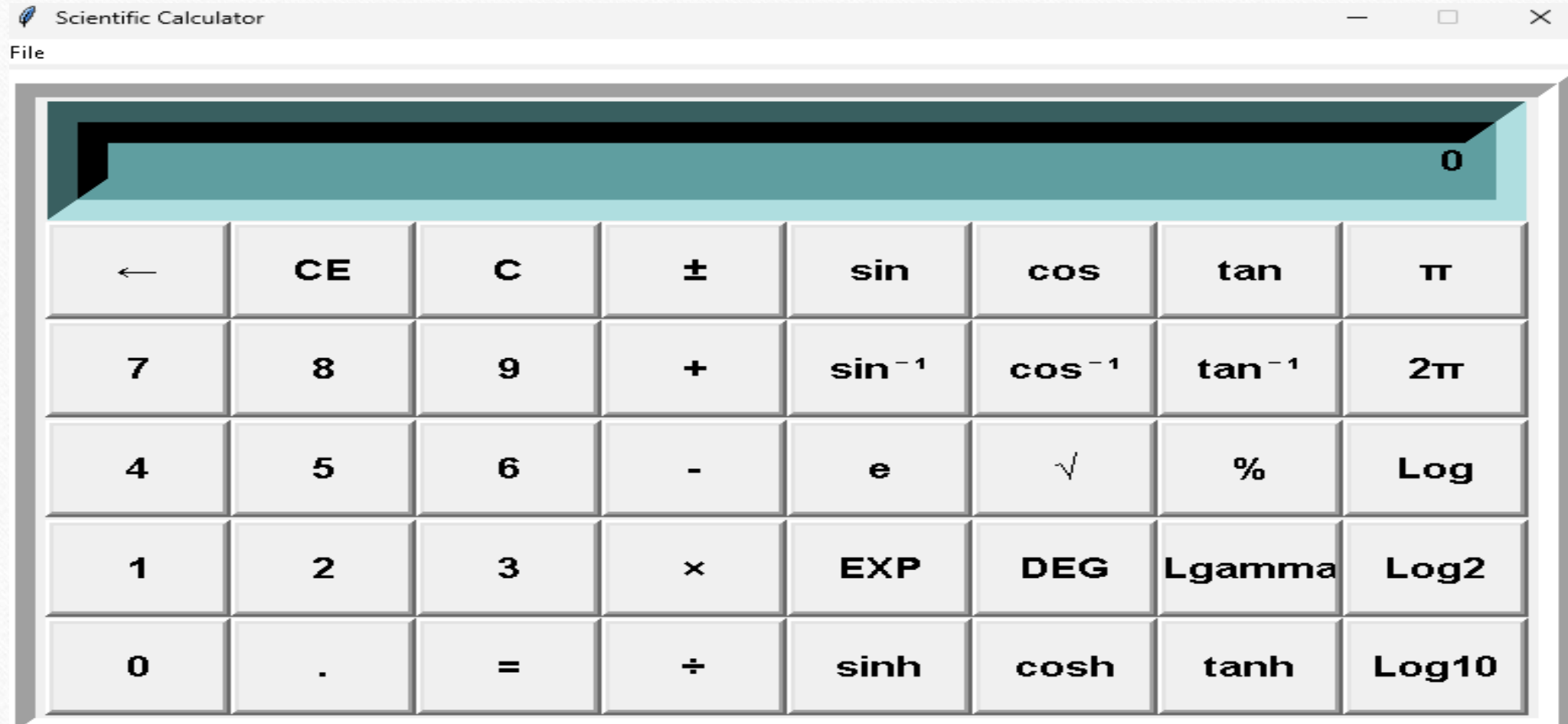
Developer Information

- Name: Yogesh Kumar
- Organization: NextHikes IT Solutions
- Department: Data Science & Artificial Intelligence

Introduction

- This project is a fully functional GUI-based Scientific Calculator.
- It supports both Standard and Scientific modes with advanced mathematical features.
- Developed using Python Tkinter library.
- Supports Standard and Scientific modes.

Application Output



Key Features

- User-friendly GUI
- Standard arithmetic operations
- Scientific functions (sin, cos, tan, log, sqrt, pi, tau)
- Error handling & clean UI
- Error handling (division by zero)

Code Structure (Part 1)

- Import libraries: Tkinter, Math
- Initialize main window
- Create frames for layout
- Setup display entry box
- Define main Frame and Calculator Frame

Code Structure (Part 2)

- Calc Class handles all mathematics
- Tracks: total, current, operator, input status
- Functions for add, subtract, multiply, divide
- Handles exceptions like divide by zero

Scientific Functions

- Trigonometric: $\sin$, $\cos$, $\tan$
- Inverse Trigonometric: $\arcsin$, $\arccos$, $\arctan$
- Logarithmic: $\log$, $\log_2$, $\log_{10}$
- Constants: π , τ , e
- Hyperbolic: $\sinh$, $\cosh$, $\tanh$

Working Mechanism

- I. User clicks a button
- II. Input processed through Calc class
- III. Operation verified and executed
- IV. Result displayed in output screen

Advantages

- Easy-to-use interface
- Covers scientific and standard functionalities
- Built with Python, highly customizable
- Lightweight application
- Start → Button Press → Check Input Type →
- Process Number / Operation → Update Display → Show Output → End

Conclusion

- This Scientific Calculator demonstrates GUI development in Python.
- It combines OOP, Tkinter, and math functions into one powerful tool.